

"آرمین بزرگری ۴۰۱۵۱۲۵۲" "میدن لیم تحلیل" "دکتر اونی"

اینجانب آرمین بزرگری متخصص لیم در این آزمون؛
از معدنی مصدقی یا لیم فنی استفاده نکنم.

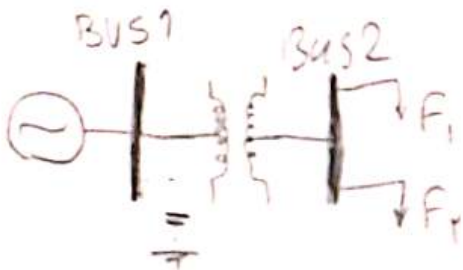
مکشی اطلاعات ندم.

آرمین بزرگری

تنها عملکرد و موضوعی عدم راد برده درج کنم.

۱۱/۵/۱۳۹۵

سوال ۱: الف)



کوره شماره ۱: $S_1 = 9 \text{ MVA}$
 $\cos(\theta_1) = 0.4 \text{ lagging}$

کوره شماره ۲: $S_2 = 11 \text{ MVA}$
 $\cos(\theta_2) = 0.5 \text{ lagging}$

$$\rightarrow \sin(\theta_1) = \sqrt{1 - \cos^2(\theta_1)} = \sqrt{1 - 0.4^2} = 0.8 \Rightarrow P_1 = S_1 \cos(\theta_1) = 9 \times 0.4 = 3.6 \text{ MW}$$

$$\rightarrow Q_1 = S_1 \sin(\theta_1) = 9 \times 0.8 = 7.2 \text{ MVAR} \quad \sin(\theta_2) = \sqrt{1 - \cos^2(\theta_2)} = \sqrt{1 - 0.5^2} = \sqrt{\frac{3}{4}} = \frac{\sqrt{3}}{2}$$

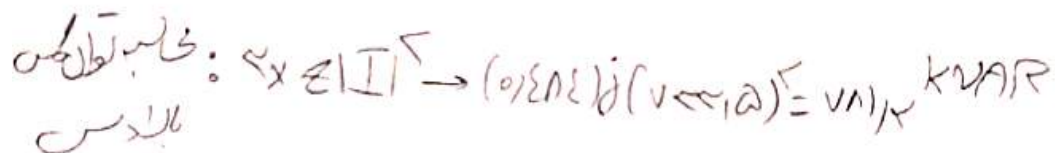
$$\rightarrow P_2 = S_2 \cos(\theta_2) = 11 \times 0.5 = 5.5 \text{ MW} \rightarrow Q_2 = S_2 \sin(\theta_2) = 11 \times \frac{\sqrt{3}}{2} = 9.526 \text{ MVAR}$$

$$P_{\text{total}} = P_1 + P_2 = 3.6 + 5.5 = 9.1 \text{ MW} \quad Q_{\text{total}} = Q_1 + Q_2 = 7.2 + 9.526 = 16.726 \text{ MVAR}$$

$$\rightarrow S_{\text{total}} = \sqrt{P_{\text{total}}^2 + Q_{\text{total}}^2} = \sqrt{9.1^2 + 16.726^2} = \sqrt{82.81 + 280.0} = \sqrt{362.81} = 19.05 \text{ MVA}$$

۱۱ "ارمن بزرگ ۵۲۱۰۴۵۲"

$$V_{rms} = \frac{V}{\sqrt{2}} = \frac{(11 \times 10^3)^2}{100} = 11000 \Omega \rightarrow \text{امپدانس} \rightarrow \frac{1}{100} \times 11000 = 110 \Omega$$



$$P.F. = \frac{P}{S} = \frac{V_1 I_1}{\sqrt{V_1^2 I_1^2 + V_2^2 I_2^2}} = \boxed{0.9214 \text{ ; } 6.2^\circ}$$



حرفاز CAF_4 و VR

$$V_1 - (V_{R1} \cos \theta_1 + V_{R2} \sin \theta_1) = \frac{V_{R2}}{\sin \theta_1} \rightarrow V_1 = 44 \angle 102.5^\circ / 101^\circ$$

$$= 44 \angle 101.89^\circ = 9.9 \angle 1^\circ$$

$\rightarrow 2AF4 = 200,109,84 \text{ C} \rightarrow \text{C} = 1,845 \text{ MF}$

"تجزیه و تحلیل مدار"



load $\rightarrow 10k + j10k$

$$I = \frac{(100 \angle 0^\circ)}{10k + j10k} = 7.07 \angle -45^\circ$$

حاصل شود: $1/4 + j1/2$

$\rightarrow (0.1 \angle 45^\circ \times 100 \angle 0^\circ) = 10 \angle 45^\circ$ $\rightarrow V_L = 450 \angle 120^\circ + (10 \angle 45^\circ \times 10 \angle 45^\circ) = 10 \angle 45^\circ$

$Q_{new} = (450 \angle 120^\circ) \times (10 \angle 45^\circ) = 4.5 \angle 165^\circ$

$Q_{old} = 10 \angle 45^\circ + 10 \angle 45^\circ = 20 \angle 45^\circ$ $\rightarrow P.F. = \frac{1}{\sqrt{1 + 1.095}} = \frac{1}{1.45} = 0.69$

$R_{ac} = 0.1 \text{ km} \frac{\Omega}{\text{mile}}$ (دوره اول)، $L = 50 \text{ km}$ (دوره دوم)

$\Rightarrow R_{old} = 0.1 \text{ km} \frac{\Omega}{\text{mile}} \times 100 \text{ mile} = 10 \Omega$

$\Rightarrow R_{line} = 10 \Omega$ $\rightarrow R_{GCM} = 2 \times 10 \Omega = 20 \Omega$

$D_{eq} = \sqrt[3]{D_{ab} D_{ac} D_{bc}} = \sqrt[3]{10 \times 10 \times 10} = 10 \sqrt[3]{1} = 10 \text{ m}$

GMR $\text{دوره اول} = 0.1 \text{ V} \times \frac{1 \text{ in}}{1 \text{ ft}} \times \frac{1 \text{ m}}{1 \text{ in}} \times \frac{1 \text{ m}}{1 \text{ cm}} = 0.0118 \text{ m}$

$GMR_{old} = \sqrt[3]{1 \times 1 \times 1} = 1$

از بین برداری $\epsilon_0 \mu_0 \omega^2$

$$L = \epsilon_0 \times 10^{-9} \ln \frac{Deq}{GMR} = \epsilon_0 \times 10^{-9} \ln \left(\frac{11190}{0.548 \times 10^{-4}} \right) = 0.1295 \times 10^{-4} \frac{H}{m} = 0.1295 \frac{mH}{km} \times 250 km$$

$$X = 2\pi f L = 2\pi \times 40 \times 0.1295 \times 250 H = \boxed{1.199 \Omega} = \boxed{0.1507 \text{ pu}}$$

بدون اثر زمین:

$$Deq = Deq_{\text{بدون}} = 11190^m \Rightarrow GMR \sqrt[4]{(2 \times 40 \times 10^6)} = (0.1295 \times 40 \times 10^6)^{\frac{1}{4}} = 0.1295^m$$

$$\gamma = \frac{d}{r} = \frac{10.1}{r} = 0.1295 \frac{m}{cm} \times \frac{250 km}{1.199} \times \frac{1}{cm} = 0.1295^m \Rightarrow C = \frac{2\pi \epsilon_0}{\ln \left(\frac{Deq}{GMR} \right) \ln \left(\frac{11190}{0.548} \right)} = \frac{2\pi \times 10^{-12}}{\ln \left(\frac{11190}{0.548} \right) \ln \left(\frac{11190}{0.548} \right)}$$

$$Y_1 = j\omega C = j \times 2\pi \times 40 \times 0.1295 \text{ MF} = \boxed{j 1.295 \text{ mS}}$$

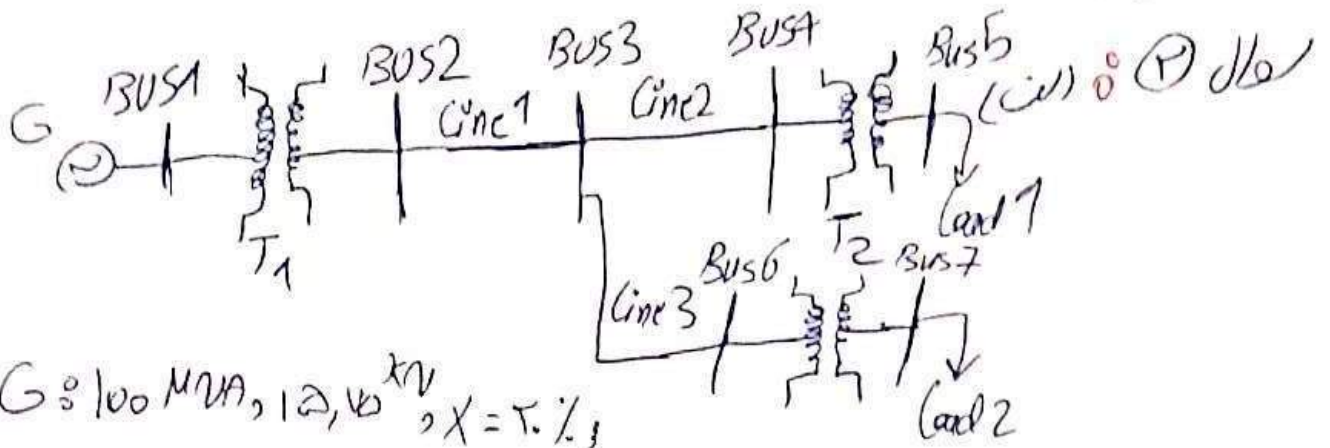
$$\approx 1.295 \times 10^{-5} \frac{F}{m} \rightarrow 1.295 \times 10^{-5} \frac{F}{km} \times 250 km = \boxed{3.24 \text{ MF}}$$

بدون زمین:

$$C = \frac{2\pi \epsilon_0}{\ln \left(\frac{Deq}{GMR} \right) - \ln \left(\frac{H_1}{H_2} \right)} = \frac{2\pi \times 10^{-12} \frac{F}{m}}{\ln \left(\frac{11190}{0.548} \right) - \ln \left(\frac{15.1}{1.0} \right)} = 1.295 \times 10^{-5} \frac{F}{km} \times 250 km = \boxed{3.24 \text{ MF}}$$

$$\left. \begin{aligned} H_{ab}' &= \sqrt{10^5 + 10^5} \\ H_{bc}' &= \sqrt{10^5 + 10^5} \\ H_{ca}' &= \sqrt{10^5 + 10^5} \end{aligned} \right\} \begin{aligned} H_1 &= \sqrt{H_{ab}' \times H_{bc}' \times H_{ca}'} = 15.1 V_0 \\ H_2 &= \sqrt{\frac{H_{ab}'}{10^5} \times \frac{H_{bc}'}{10^5} \times \frac{H_{ca}'}{10^5}} = 1.0^m \end{aligned}$$

$$Y = j\omega C = j 2\pi \times 40 \times (3.24 \times 10^{-6}) = \boxed{j 1.295 \text{ mS}}$$



$G: 100 \text{ MVA}, 12.5 \text{ kV}, X = 2\%$

$T_1, T_2: 20 \text{ MVA}, 12.5 \text{ kV} / 10 \text{ kV}, X = 10\%$

$T_1, T_2: 100 \text{ MVA}, 16 \text{ kV} / 250 \text{ kV}, X = 5\%$

$\text{Line1, Line2, Line3}: X = 2\%$

$\text{Load1} (20 \text{ MVA}, 10 \text{ kV}, \text{PF} = 0.9 \text{ lagging})$
 $\text{Load2} (10 \text{ MVA}, 250 \text{ kV}, \text{PF} = 0.8 \text{ lagging})$

$V_{\text{base}} \rightarrow V_a = V_b = 10 \text{ kV}, V_c = V_d = 250 \text{ kV}, V_1 = 19 \text{ kV}$

$X_G = X \times \left(\frac{V_{\text{old}}}{V_{\text{new}}} \right)^2 \times \left(\frac{S_{\text{new}}}{S_{\text{old}}} \right) = 0.2 \times \left(\frac{12.5}{19} \right)^2 = 0.145 \text{ pu} = X_G$

$X_{T_1} = 0.1 \text{ pu}$
 $X_{T_2, T_1} = X \times \frac{S_{\text{new}}}{S_{\text{old}}} = 0.1 \times \frac{100}{20} = 0.5 \text{ pu} = X_{T_2} = X_{T_1}$

$\text{Load1} \rightarrow \frac{20}{10} \angle \cos^{-1}(0.9) = 0.2 \text{ pu} + j 0.13 \text{ pu} = S_{\text{Load1}}$

$\text{Load2} \rightarrow Z_{\text{base}} = \frac{V_b^2}{S_b} = \frac{10^2}{100} = 0.1 \Omega \rightarrow Z_{\text{Load}} = \frac{Z}{0.1} = 10 + j 13 \text{ pu} = Z_{\text{Load}}$

